Learning Resource — Understanding the Thesis

> Title: Cooperative Game Theory for Explainable Artificial Intelligence (XAI) in Recommendation Systems: A Shapley Framework for Actionable Insight.
>
> Author: Mouad Louhichi · Supervisor: Pr. Mohamed Lazaar
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> Institution: ENSIAS, Mohammed V University, Rabat, Morocco.

This resource teaches you the thesis from first principles. It is written so that a colleague, a student, or a jury member can (a) understand the argument, (b) follow the method, and (c) defend its claims. It is organised as five modules that mirror the defense deck: the core idea, the mathematics, the three contributions, the protocol, and the honest boundaries.

Module 0 — The thesis in one sentence

> Shapley values (cooperative-game attribution) are used not only as post-hoc explanations but as a single, shared, formally-grounded mechanism that (1) explains black-box clustering, (2) stays coherent when that explanation is scaled to large, hierarchical, multi-level clustering, and (3) becomes an in-training optimisation signal inside a dynamic hypergraph recommender (DyHuCoG).

Three words to remember: explain (C1), scale (C2), integrate (C3).

Module 1 — Why this problem matters

1.1 The core tension

Recommendation and clustering systems grew more expressive over time — from similarity filters, to matrix factorisation, to neural collaborative filtering, to graph GNNs, to hypergraph models. Every step improved ranking but lowered transparency. The thesis keeps accuracy and interpretability as objectives to be reconciled, not traded against one another.

1.2 Definition 1.1 — Actionable insight

An explanation is actionable when it:

1. **Identifies at least one modifiable factor whose change is associated with a**

specifiable change in model output, and

2. **That factor is expressible in the semantic vocabulary of the task domain — a**

physicochemical variable (wine), a pollution indicator (air quality), or a preference signal (recommendation) — not an opaque latent code.

> Why this matters: an explanation that identifies a modifiable driver supports intervention, not merely description. A credible explanation is judged by whether it supports a downstream decision, not by whether it looks plausible.

1.3 Why the deficit matters

- Trust — opaque systems mediate what billions of users see, buy, and watch.
- Debugging & science — you cannot learn from a model you cannot interrogate.
- Regulation — the EU AI Act (Reg. 2024/1689), OECD AI principles, and GDPR all put a

premium on meaningful explanation.

Module 2 — The mathematical backbone

2.1 Cooperative (transferable-utility) game

A TU game is a pair (N, v) where N is a set of players and $v: 2^N \to \mathbb{R}$ is a coalition value function. Intuitively, $v(S)$ answers: how valuable is the coalition S?

2.2 The Shapley value

For a player i, the Shapley value is the expected marginal contribution of i over all orderings of the players:

$$\varphi_i(v) = \Sigma_{\{S \subseteq N\setminus\{i\}\}} \left[|S|!\,(|N|-|S|-1)! \,/\, |N|! \right] \cdot \left[v(S \cup \{i\}) - v(S) \right]$$

It is the unique allocation satisfying four axioms:

Axiom	Formal statement	Interpretive meaning
Efficiency	$\Sigma \varphi_j = v(N) - v(\varnothing)$	All explanatory mass is allocated
Symmetry	Identical marginal contributors get equal allocation	Equal players are treated fairly
Null player	Zero marginal contributor gets zero	No spurious credit
Additivity	$\varphi(v + w) = \varphi(v) + \varphi(w)$	Explanations compose across tasks

> Why Shapley over LIME? SHAP is grounded in an axiomatic, normative basis; LIME fits a
> local surrogate and has no equivalent guarantee, and it is sensitive to perturbation design.

2.3 Approximation — the practical necessity

Exact Shapley is exponential. The thesis uses two approximations:

- Monte Carlo estimator (used in DyHuCoG):

$\hat{\varphi}_j = (1/M)\ \Sigma_m [\ v(S_m \cup \{j\}) - v(S_m)\ ]$. Unbiased; variance $= \sigma^2/M$; MSE $\to O(1/M)$; absolute error $\to O(1/\sqrt{M})$.

- TreeSHAP (used in the clustering chapters): exact/fast attribution for tree models, which is why a LightGBM surrogate is chosen.

2.4 The key caveat (be ready to state it)

In the surrogate pipeline, efficiency holds with respect to the LightGBM surrogate output, not directly to the Silhouette-based game. That is exactly why surrogate fidelity is the critical validity condition.

Module 3 — The three contributions

C1 — Explainable black-box clustering (Chapter 5; Paper I)

Goal: explain a clustering (an unsupervised partition) faithfully and actionably.

The gap: explainable clustering is fragmented — methods favour a local or a global explanation, rarely both; they often fail to scale; and they rarely preserve coherence across

clusters. Shapley is well established in supervised tasks but almost absent from unsupervised
clustering.

The method — cooperative game framing. Players are features (N = F). The coalition value is the clustering quality of a feature subset:

v(S) = Silhouette(KMeans(XS, k))

Silhouette is chosen because it is bounded, normalised, and semantically intuitive. The
problem: evaluating v(S) for every subset is combinatorial. Bridge: train a LightGBM multiclass surrogate to predict the induced cluster labels from the original features, then apply TreeSHAP to the surrogate. This keeps attribution in the original semantic feature space.

Pipeline (5 stages): standardise → PCA (diagnostic only) → K-Means++ + optimal-k → LightGBM surrogate → TreeSHAP.

Key result (Wine, 4,898 × 11):

- k is chosen by interpretability, not geometry. k=2 gives Silhouette 0.214 / DB 1.775;

selected k=3 gives Silhouette 0.144 / DB 2.097 — weaker separation but three distinct
oenological narratives.

- Global importance: density → pH → fixed acidity → sulfur-dioxide → alcohol.
- Cluster profiles: C0 ~ density + SO_2; C1 ~ acidity/pH; C2 ~ acidity, alcohol, related.
- Surrogate fidelity: macro-F1 ≈ 0.82 (the floor).

Findings: Shapley explains black-box clustering faithfully and coherently; it returns attribution to original variables; it is theoretically grounded (four axioms); it recovers a
chemically interpretable hierarchy.

Limitations: depends on LightGBM surrogate fidelity; tabular only; single-level (cannot explain how importance reconfigures between a partition and its sub-partitions).

Takeaway: Shapley attribution is a single principled lens for explaining an

unsupervised
partition — but real data are rarely single-level. This motivates C2.

C2 — Multi-level XAI for large-scale clustering (Chapter 6; Paper II)

Goal: keep the explanation coherent when clustering becomes multi-level / large-scale.

The gap: a variable can be globally important yet locally uninformative (or the reverse). A flat explanation is true yet incomplete — it cannot show how importance changes as you "zoom in."

The method. Recursive/nested clustering: coarse clustering on the full dataset, then subdivide each cluster. For each level, train a level-specific surrogate and compute SHAP in
the same original feature space. Cross-level aggregation is NOT a naive average — it respects cluster size and nesting structure.

Proposition 6.1 — Hierarchical Attribution Consistency. Let Φ^(ℓ,c)j = E{x~c}[|φj^(ℓ)(x)|]
be the expected absolute SHAP importance of feature j at level ℓ in cluster c, and let w{c'} = |c'|/|c| be the relative child size. For a strict nested hierarchy on a consistent feature space:

Φ^(ℓ,c)j = Σ{c' ∈ child(c)} w{c'} · Φ^(ℓ+1,c')j + εj

where εj is a residual from surrogate mismatch, vanishing under perfect fidelity. Derived
via the law of total expectation (children partition the parent).

> It does NOT claim explanations are identical across levels; it implies differences can be interpreted rather than dismissed as inconsistency.

Key result (Beijing, 383,585 × 11):

• Full dataset, k=3; Silhouette ≈ 0.63, Davies–Bouldin ≈ 0.55 (stronger than wine — do not confuse this with wine's 0.144/2.097).

• Global importance: temperature → dew point → pressure → CO → NO_2 → PM10 → PM2.5.

Meteorological variables play a structurally central role (they condition dispersion, trapping, photochemistry).

- Three regimes: warm photochemical (ozone, temp, dew point); wintertime smog (CO, SO_2,

PM, low wind); clean/well-dispersed events.

- Multi-level insight: at coarse level temp/dew point dominate (regime selection); within

clusters CO, SO_2, PM10, wind speed, pressure, ozone become discriminative (variation within regime). This is the point of a multi-level explanation — not a contradiction.

Comparative: Beijing Silhouette 0.63 vs Gramegna & Giudici (credit-risk) 0.37; LIME shows weaker structural coherence and less stable local narratives.

Findings: scalable multi-granular explanation; formal cross-level consistency (Prop 6.1); validated on a structurally different dataset.

Limitations: clustering remains static (despite temporal Beijing data); surrogate + representative-instance reporting compress observation-level variation; tabular only; still an explanation of a pre-computed partition.

Takeaway: Shapley attribution stays coherent across granularity when the hierarchy is explicit — but it is still post-hoc. This motivates C3.

C3 — DyHuCoG: Dynamic Hypergraph Cooperative Game (Chapter 7; Paper III)

Goal (the strongest claim): move attribution beyond post-hoc analysis into the learning dynamics of a recommender — attribution becomes an in-training signal.

The gap: graph/hypergraph recommenders treat message importance as either uniform or attention-weighted, without a principled marginal-contribution account. Diversity is a secondary objective or a re-ranking heuristic. Interpretability is added after prediction.

Cooperative-game framing. Players $N = U \cup I \cup C$ (users, items, contexts). Hypergraph $H = (V, E, W)$. Coalition value (multi-objective utility):

$$v(S) = \alpha \cdot \mathrm{NDCG@20}(S) + \beta \cdot \mathrm{Diversity}(S) + \gamma \cdot \mathrm{ContextScore}(S), \quad \alpha+\beta+\gamma=1$$

$$v_{pref}(S) = v(S) + \lambda_{pref} \cdot \Sigma_{(u,i)\in S}\, \mathrm{sim}(u,i)$$

Tuned: $\alpha=0.60$, $\beta=0.25$, $\gamma=0.15$, $\lambda_{pref}=0.20$. Coalition evaluation is scoped to the interaction episode (a few dozen players), not the full catalogue — an honest boundary.

Preference-aware Monte Carlo Shapley. $\hat{\varphi}_j^{pref} = (1/M)\ \Sigma_m\ [\ v_{pref}(S_m \cup \{j\}) - v_{pref}(S_m)\ ]$.

$M=50$ gives MSE $\approx 1.4\times10^{-5}$, ~99% accuracy. Refreshed every 10 batches, smoothed by EMA.

Architecture. Base propagation: $e^{(\ell+1)} = \sigma(D^{-1/2} A D^{-1/2} e^{(\ell)})$. Shapley-weighted:

$e_j^{(\ell+1)} = \sigma(W^{(\ell)} e_j^{(\ell)} + \Sigma_{k\in N(j)} w_{jk} e_k^{(\ell)})$ with normalised weights

$w_{jk} = \hat{\varphi}_{jk} / \Sigma_{k'\in N(j)} \hat{\varphi}_{jk'}$ (clipped + smoothed). Layer fusion with learned α_ℓ;

attention gate $a_{ui} = \sigma(W_a[e_u, e_i, l_i])$; context-aware score

$f(u,i,c) = y_{ui} + \lambda_c \langle g(c_{ui}), e_{cui}\rangle$.

Loss. $L = L_{rec} + \lambda_{div} L_{div} + \lambda_{ctx} L_{ctx} + \lambda_{reg} L_{reg}$ — BPR ranking, intra-list diversity regulariser, context alignment, weight decay. The learning objective and coalition value are aligned.

Key results (relative to strongest baseline HPCF):

Dataset	NDCG@20	Recall@20	Coverage	ILD
MovieLens-1M	+9.77%	+12.58%	+16.1%	+11.9%
Amazon-Book	+13.33%	+16.16%	+29.7%	+12.5%

- Largest relative gains on the sparsest data (Amazon) → Shapley weighting helps most when signal is weak.
- Ablation: removing Shapley value costs −4.6%/−6.1%; removing context is the

largest

single loss (−8.2%/−11.0%); hypergraph −6.8%/−8.9%; attention −3.5%; diversity −5.8%.

- Statistical validation: paired t-test per-user NDCG@20 ($n=6040$), vs HPCF $t=46.38$, Cohen's $d=1.33$, $p \approx 1.81\times10^{-270}$; Holm–Bonferroni + Wilcoxon signed-rank. (Paired tests apply only to MovieLens-1M; Amazon remains descriptive.)
- Cold-start: user ~0.061 (+10.9%), item ~0.057 (+9.6%).
- Computational overhead: ~1.78× training time (2000s vs 1125s); inference 1.84 ms/query (ML-1M).

Findings: attribution as an in-training signal; the accuracy–diversity–context trade-off is not structurally fixed; ranking/coverage/diversity improve together.

Limitations: ~1.78× training overhead; depends on meaningful context; MC Shapley could use variance reduction; ablation is component-wise (no factorial interactions); baselines finalised early 2026 (claim only vs tested set).

Takeaway: attribution is a first-class part of the learning objective; the explanation is a direct read-out of what the model already optimises — structurally faithful, not an external approximation.

Module 4 — The shared experimental protocol (Chapter 4)

Datasets

Dataset	Scale	Density	Role
Wine Quality (Vinho Verde)	4,898 × 11	dense, chemically correlated	C1 single-level clustering
Beijing Multi-Site Air Quality	383,585 × 11	large, noisy, 2013–2017	C2 multi-level clustering
MovieLens-1M	6,040 u × 3,706 i	0.0447 · 1.0M ratings	C3 DyHuCoG
Amazon-Book	52,643 u × 91,599 i	0.0006 · 3.0M ratings	C3 DyHuCoG

Selection criterion: clustering datasets have semantically interpretable features; recommendation datasets are benchmark-standard with established baselines.

Splitting

- Recommendation: user-level temporal holdout — 70% train / 10% val / 20% test; leave-one-out (latest test positive per user is the target).
- MovieLens-1M ratings > 3 treated as positive implicit feedback.
- Popularity-aware negative sampling $q(i) \propto f_i^{\eta}$ for harder contrasts.
- Clustering: 5-fold CV for surrogate/attribution stability.
- Reproducibility: seeds {42,43,44,45,46}; early-stopping patience 20.

Baselines

Recommendation: MF, NCF, LightGCN, RecDCL, HCCF, HPCF (strongest). Clustering: LIME-based
surrogate pipeline.

Metrics

- Ranking: Precision@K, Recall@K, NDCG@K (NDCG@20 primary; $DCG = \Sigma (2^{rel}-1)/\log_2(i+1)$).
- System diversity: Catalogue Coverage = $|\cup R_u| / |I|$.
- List diversity: $ILD(R_u) = 2/(K(K-1)) \Sigma_{a<b} [1 - sim(i_a, i_b)]$ in the learned item space.
- Clustering: Silhouette, Davies–Bouldin, Calinski–Harabasz.
- Statistics: paired t-test, Holm–Bonferroni, Wilcoxon signed-rank, Cohen's d_z.

Hardware / software

Intel i9-14900K (24 cores) · RTX 4090 24GB · 48GB RAM · 2TB SSD · Python 3.8 · scikit-learn / LightGBM / SHAP / PyTorch 2.0.1 · Altair for interactive SHAP.

Module 5 — The five research questions & how they map

RQ	Question	Answered by
RQ1	How can Shapley explain black-box clustering faithfully at instance and cluster level?	C1 (Ch. 5)

| RQ2 | How can this extend to large-scale, hierarchical clustering without losing tractability or consistency? | C2 (Ch. 6) |

| RQ3 | Can cooperative attribution move beyond post-hoc and enter the learning dynamics of graph recommenders? | C3 (Ch. 7) |

| RQ4 | Can a recommender jointly optimise ranking, context and diversity when importance is estimated by a cooperative-game utility? | C3 (Ch. 7) |

| RQ5 | What emerges when clustering explanation and recommendation learning are two stages of one cooperative-game perspective? | Thesis-level (Ch. 8) |

Thesis answer: cooperative game theory can function as a shared methodological

perspective for actionable explanation across clustering and recommendation. It is not

claimed to be one fully unified framework eliminating all tension; it is a common

attribution language.

Module 6 — Honest boundaries (be ready to concede these)

1. **Computational: exact Shapley is intractable; every contribution relies on**

approximation, surrogates, or restricted reporting.

2. **Methodological: clustering depends on surrogate fidelity; recommendation depends on**

stable approximate contributions and adequate context.

3. **Empirical: tabular clustering + offline recommendation; no multimodal / sequential /**

online deployment; no dedicated human-subject actionability study.

4. **Claim scope: a coherent perspective, not one fully unified framework.**

5. **Efficiency scope: holds w.r.t. the surrogate output, not the Silhouette-based game.**

6. **Scoped player set (DyHuCoG): per-episode valuation, not the full catalogue.**

7. **Baselines capped at early 2026: superiority claimed only vs the tested set.**

Module 7 — The one central claim worth defending

Explanation should be part of the modelling logic itself, not a by-product attached after

prediction. Shapley-value attribution — because it satisfies efficiency, symmetry,

null-player, and additivity — provides a normative (not heuristic) basis for allocating

explanatory responsibility across features, interactions, and contexts.

Quick recall checklist

- [ ] What is an actionable insight? (modifiable factor + domain vocabulary, not a latent code)
- [ ] What are the four Shapley axioms? (efficiency, symmetry, null player, additivity)
- [ ] Why a surrogate for clustering? (TreeSHAP explains trees, not centroids)
- [ ] What does Prop. 6.1 guarantee? (cross-level differences are interpretable, not noise)
- [ ] How does DyHuCoG make attribution an in-training signal? (Shapley-weighted message passing)
- [ ] What is the accuracy–diversity gain pattern? (both improve together; largest on sparsest data)
- [ ] What are the honest limits? (overhead, surrogate fidelity, offline/tabular, early-2026 baselines)